



MyCelia

Group: WAJI (Will, Aiden, John, Ibraheem)
Client: Urban Forestry of Lancaster



Motivation

Our Client (Mike Meyer) wanted to:

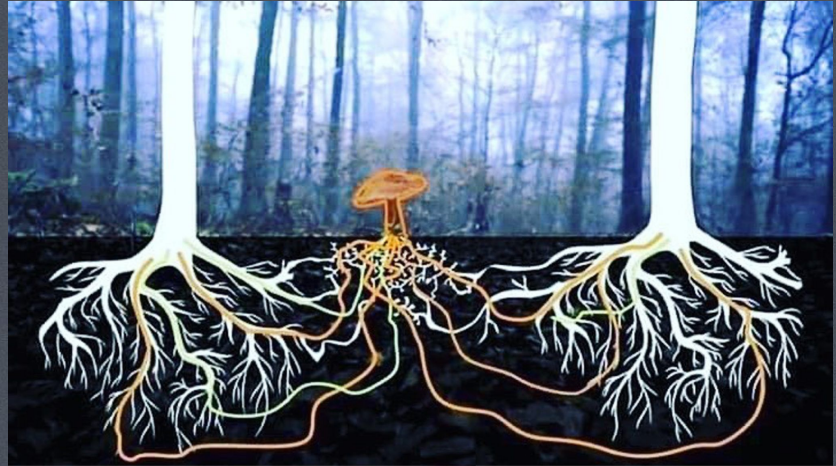
- Increase public engagement with urban trees
- Make tree data accessible and engaging
- Foster community connection with urban nature
- Support education in real-world settings



What does MyCelia mean?

A play on the mycelium roots (aka the mycorrhizal networks) that connect trees in a healthy forest.

We found this fitting, as the app is meant to connect people through games, exploration and social media features.



MVP Structure

Three core ideas: **learning**, **exploration through gameplay**, and **social interaction**

Exploration

Map Interface

Tree Dataset Integration

Tree Information System

Gameplay

Game Loop

Goals & Narrative

Achievements

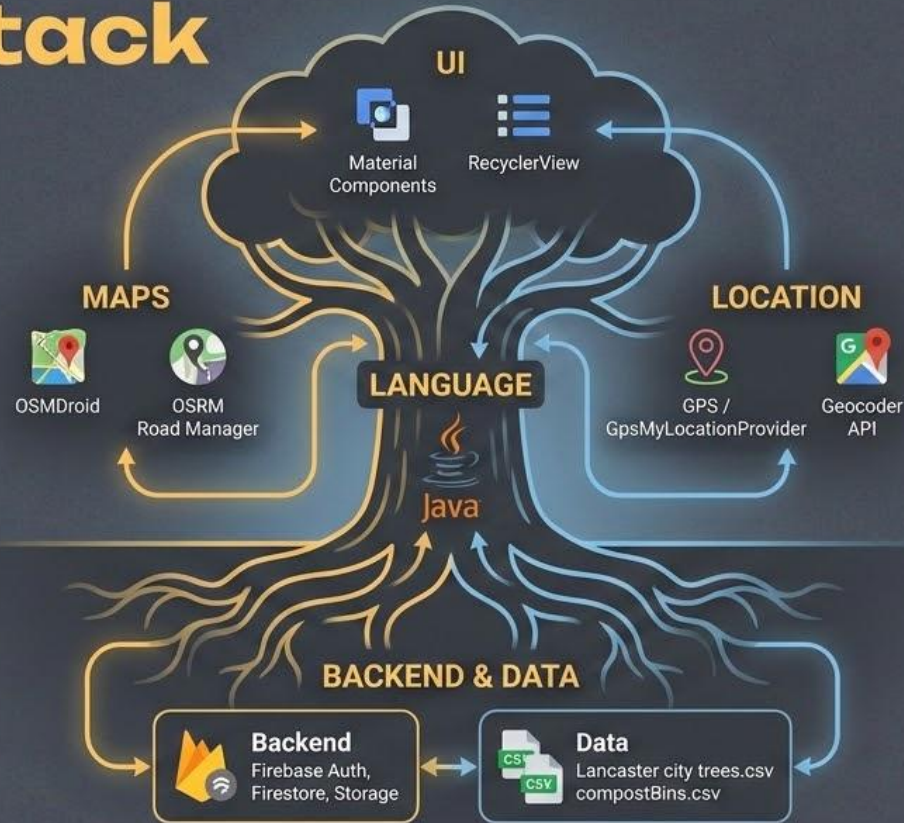
Social System

User Profiles

Posts (Images + Text)

Full-Stack Integration
with a Firebase backend

Tech Stack



Authentication

urban-forestry ▾

Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

i The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps. ▾

🔍 Search by email address, phone number, or user UID Add user ↺ ⋮

Identifier	Providers	Created ↓	Signed In	User UID
mdnguyen123321@gm...	✉	Apr 20, 2026	Apr 21, 2026	yHvIvdYkAOhCdtDPsGXphuy...
abailley@fandm.edu	✉	Apr 20, 2026	Apr 20, 2026	GdtRhB...
jbirkela@fandm.edu	✉	Apr 16, 2026	Apr 22, 2026	WhlI8DK...
will@hotmail.com	✉	Apr 14, 2026	Apr 14, 2026	daBAXQ...
ibraa@yahoo.com	✉	Apr 14, 2026	Apr 20, 2026	m4alIL14...
ibra@yahoo.com	✉	Apr 7, 2026	Apr 7, 2026	Gynjhl8...
waji@gmail.com	✉	Mar 31, 2026	Mar 31, 2026	RSp6yDZeOIRwMyiV4AFilWZa...



firebase



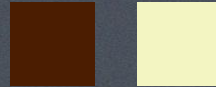
App Startup

Users can create accounts or log in with their email address

Returning users are automatically recognized and redirected to the main map interface



Logos & Seasonal Themes



About Page

- Provides acknowledgements to important people in the Urban Forestry department
- Promoting Lancaster Urban Forestry Department and their role in supporting the urban canopy and inspiring the project.
- It also provides links for users to access the app's social media links



A Live GPS Tree Map

Tree Coverage (Current Scope)

The app is currently focused on trees in Buchanan Park, with nearly 60 different species. This allows us to validate core features in a controlled but diverse environment before expanding to other locations.

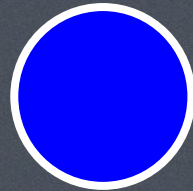
Buchanan Park is next to the North Museum, which Urban Forestry plans to collaborate with for educational programming and outreach.

Future possibilities

While the app is currently restricted to Buchanan Park, it is designed to be scalable so that future versions could cover all of Lancaster or an even larger area.

Map Markers

- Trees (from a csv of Lancaster Urban Forestry's list of trees)
 - Eventually will be changed to reading directly from the sheet
- F&M Compost bins
 - Also from a spreadsheet provided by Professor Eve Bratman
- User photos and their locations

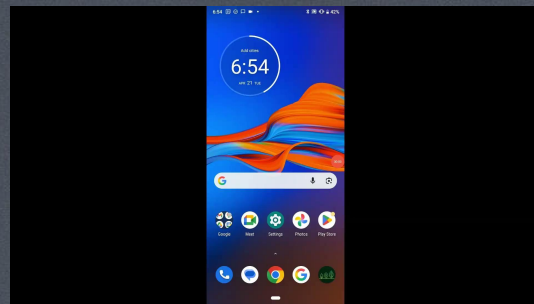


Game Loop

Purpose: To **encourage people to explore** the urban canopy and **educate them about trees**

Our original idea was to make a Pokemon-Go style game where users explore the neighborhood trees and collect some sort of achievements

[DEMO link](#)





How to Play

- Users are randomly assigned two goals at a time
 - To make progress, players must approach different trees in the park and open their info pages in the app.
 - This feature provides variety and an incentive to learn about the various trees in the area
- 



Achievements

- Ideas were brainstormed to encourage exploration
 - Exploring different species
 - Exploring different characteristics
 - Once you complete a goal, your achievement associated with that goal ticks up. If you complete a certain goal multiple times you'll level up in that area.
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Community Feed - Front End

- Post (Image/Text)
- Like system
- Comment system
- Report system
- Edit Post Caption
- Remove Post

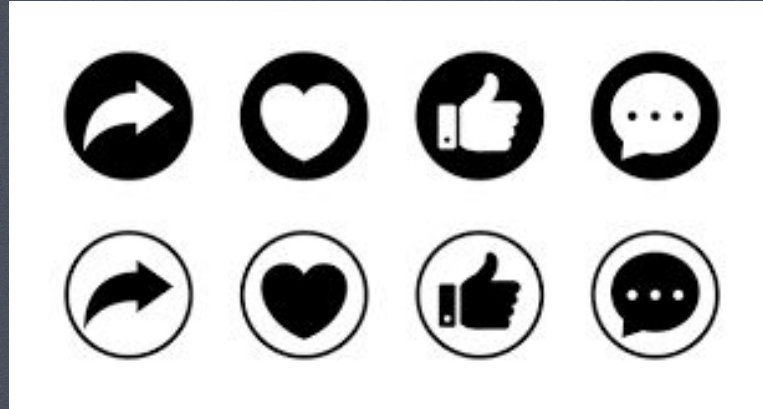




Image-to-Location Navigation

Interest Sharing

Users can post their favorite tree photos and have the option to assign a location along with the image

Community Interaction

Users can click on a shared image to view the exact location where it was taken. The app then guides them to that spot on the map





User Profiles

Shows:

username, full name, total # posts, total # likes

Upload/Change Profile Picture

Edit your information



Community Feed - Database

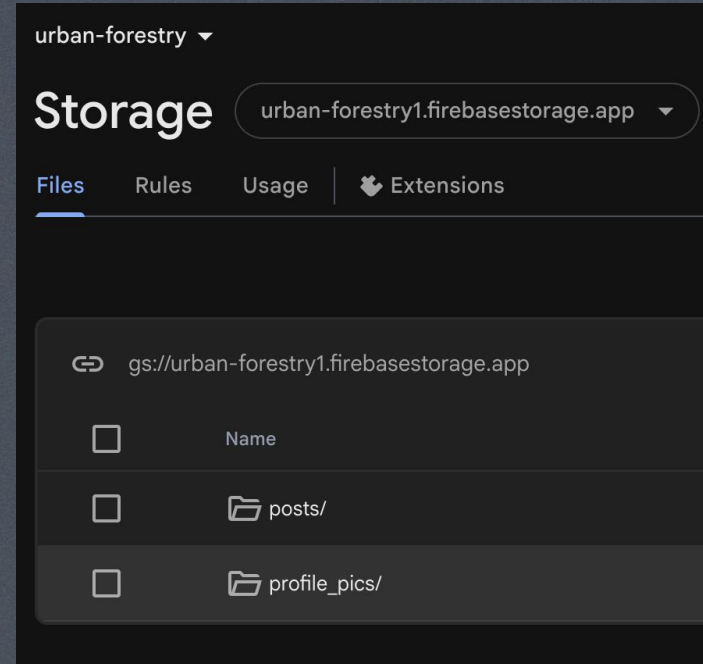
FIRESTORE

(Structured
Data)

🏠 > posts > Vi6PPouOInp87... ✎		
📁 (default)	📁 posts	📄 Vi6PPouOInp87xpM6IBX
+ Start collection	+ Add document	+ Start collection
posts >	4HhLLnP6CHwKLizpR9nt	comments
users	8nVcu9S8HxLHcQNXVvwy	likes
	MZmOLidv3fjAGaxbJmGF	
	0iPmoQofgPEqmdjhJoU0	+ Add field
	P1RMNEFRxEVD0v5BB9iE	caption: "I spruced on my spring till it steened"
	Qzh9RHJSRgEj2neZzyV7	commentCount: 1
	⋮ Vi6PPouOInp87xpM6IBX >	createdAt: April 21, 2026 at 7:07:03 PM UTC-4
	XXdoFQQkmTixKEmpW1up	hasLocation: true
	b078S0tRYT9s8kPVgGRi	imageUrl: "https://firebasestorage.googleapis.com/v0/b/urban-forestry1.firebaseio.com/o/posts%2Fdf8e36c4-f9ff-4c99-982c-62c83181f4fa.jpg?alt=media&token=36dcf71d-b1c9-4ecf-bcaf-e991fb4e424d"
	ndH9nxtmmo1gC1F8ahM3	latitude: 40.047173
	tgZLo17BJNm8kBUuoGto	likeCount: 1
		longitude: -76.3224631

Community Feed - Database

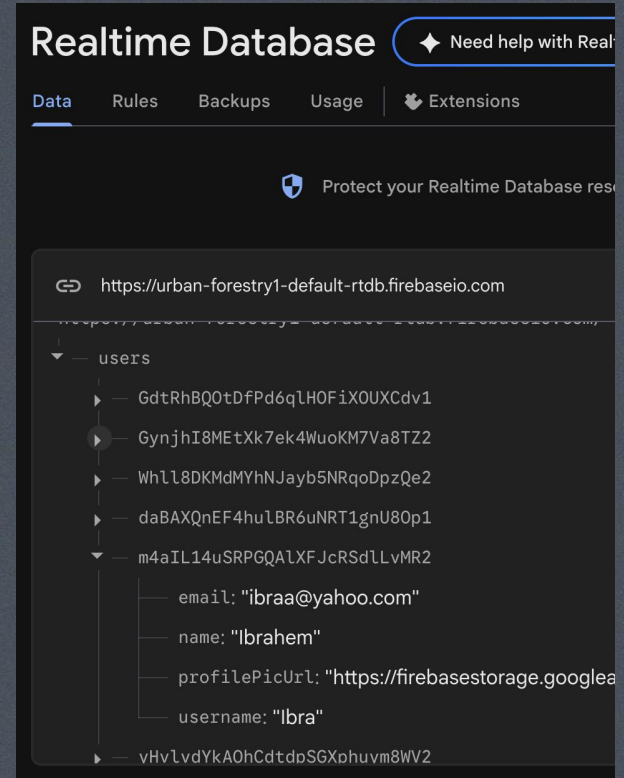
FIREBASE STORAGE (IMAGES)



Community Feed - Database

Realtime Database:

- Storing profile data for quick fetch (low latency)



Quality Assurance

Given the app's several intertwined features, we wanted to make sure that the user experience was smooth and straightforward.

Our process included:

- Look at the app from the perspective of a new user; search for potential sources of confusion or frustration
- Use consistent styling for an integrated feel
- Test every feature for bugs and ensure the app doesn't crash



Future Work

Data cleaning on Firebase

Store achievements on user account instead of shared preferences

Multilingual App

Expand the tree database beyond Buchanan Park





Thank you's

Thank you to Mike for being really supportive

Thank you to Professor Novak for always being there



Thank you all for listening!

Any questions?